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**Understanding the relationship between the Pokemon type and its battle role using the
official performance statistic : Pokémon Database, 2008-2025**

Term Paper

CSD105 Data Science

Instructor: Professor Hiral Vegda



ABSTRACT

The term paper aims to investigate the fundamental connections between a Pokemon's elemental type and its functional battle role. The goal is to find if the existing correlations are simply a coincidence or an actual deliberate design. The study's central essence is to find out if a statistically predictable relationship exists between classification of Type and the quantitative distribution of the six Base Stats (HP, Attack, Defence, Sp, Atk, Sp, Def Speed). Using the official dataset of over 1000+ Pokemons, the analysis focuses on the base statistics mentioned above. The methodology used mainly are the two statistical methods we learned in our CSD105 course itself : Analysis of Variance and correlation. Analysis of Variance is used to test if 'Type' has an actual and significant impact on the 'Base Stats'. Analysis of Variance, it basically compares the average value of every stat individually across all different Type groups. Correlation Variance, basically analyzes the relationships between individual statics and how those relationships actually may change within a Type.

INTRODUCTION

The Pokémon franchise centers on creatures, often based on animals, myths, and objects, that inhabit the Pokémon World. Humans, called Trainers, journey across regions to catch, train, and battle with these creatures, with the ultimate goal being to complete the Pokédex (an encyclopedic catalog) and become a Pokémon Master. This core loop of collecting and competitive battling drives the entire universe across games, anime, and cards. Pokémon stats exist to create unique combat roles and ensure balance—a fast, frail Pokémon is different from a slow, heavy attack one. Categories like Legendary and Mythical Pokémon denote creatures tied to the deep lore of the world, often playing roles in creation, weather, or natural forces.

In the complex and chaotic multiverse of the pokemon franchise, a battle's success is dependent upon the vast range of interconnected variables such as its type (elemental classification). However this straightforward label underlies strategic planning of those variables mentioned earlier. These variables essentially form the supposed DNA of a pokemon's combat potential and abilities. Hence a pokemon's statistical dispersion of its variables govern its function. For instance high speed and special attacks create a term known in the pokemon world as a "Special Sweeper", created to outpace the opponents. This made us wonder, whether this was all just a coincidence or a deliberate selection of certain variables or factors. Is there actually a measurable connection between the pokemon type and its battle role it was originally meant to fit?

With the functional battle roles being established by the fandom, the fundamental question remains unanswered regarding the correlation between 'Type' and 'Statistical' dispersion. Whether the relationship is merely just a coincidence, or if it is the developer's choice of character design, the term paper gives an insight on the workings of the developers' mind.

The Type label refers to the elemental classification which serves as a proxy for a Pokémon's intended statistical profile and combat strategies. This paper seeks to test this assumption itself, based on the data. Therefore, the problem statement is to determine if a statistically

accurate and predictable relationship exists between the Pokémon's Type classification and the six core Base stats that serves as the Pokémon's combat nature.

The term paper pursues three primary objectives to seek a conclusion on the same. First, the provided dataset would be substantially analyzed by us to understand the general distribution and variance of Base statistics across all Pokémon. Second, we will be using statistical methods (specifically Analysis of Variance and correlation analysis) to quantify the strength of relationship between the specific types (e.g., fairy, steel, fighting), and relative strength of the individual Base stats (e.g., defense vs attack). Lastly, we will objectively group Pokémon into battle roles based purely on their statistical abilities which will in turn, provide us with a quantitative basis for Type-Base relationship assessment. This clustering will prove if the type variable predicts a Pokémon's strategic application.

The goal is to offer an assessment backed up by data and evidence of the underlying logic design that governs the Pokémon universe. The data utilized in the study is a dataset encompassing over a thousand distinct Pokémons from the entire franchise. The analysis puts a significant focus on the Base stats which are static values and represent the measure of a Pokémon's raw potential to navigate a combat.

By isolating these six statistical variables of Type, we are essentially creating a controlled environment for our analysis by avoiding interference of game elements like abilities, natures, or complex adaptable movesets. This allows us to test the core hypothesis regarding the statistical validity of the Type assessment, ensuring that the deep-rooted structure of the creature design correlates.

The findings of this term paper holds significant implications of concepts beyond simple fan curiosity. If a strong and predictable link is established between Type-Base stat stratification, it will suggest the consistency of the game's balance design to be intentional, encouraging a model that serves as a large-scale character creation.

Conversely, if the correlation proves to be weak, it would suggest that the strategic roles might be a product of secondary variables like player behaviour or moveset access, rather than an intentional design mandate. In short, this research provides a quantitative, data based framework to understand the functional anatomy of Pokémons and how the core design dictates the variables concerning viability and strategic utility.

LITERATURE REVIEW

The Pokemon Database (2008-2025) provides the explicit classification of Type and the six core statistics has already established a Pokemon's function based on these statistics. The community defined roles are as follows:

Glass Canons: Pokemon which have high Attack or Special Attack and Speed but low Defence and HP. Their main role is to defeat an opponent in a single blow with their speed before they themselves take a hit.

Walls: Pokemon which have high HP, defence or Special Defence. Their role is to absorb the damage and stall opponents as much as possible.

Tanks: It has a hybrid role, the pokemon has high HP and high offence. It is designed to both deal and handle damage.

Sweepers: These are fast offensive Pokemons intended to literally 'sweep' an opponents team at once when their defences are weakened.

Pivots: High speed and access to specific moves although the overall statistic is balanced, allowing them to switch in and out of the battle.

The Pokemon dataset being one of the most popular dataset for introductory data science, the most common form of analysis uses descriptive statistics to find relationships between Type and the six core stats. The previous studies done on kaggle have constantly shown that a Pokemon's Type is a strong predictor of its stat bias. The findings are as follows:

Fighting Type: These Pokemon have a statistical higher median Attack stat.

Psychic Type: These Pokemon have shown a clear bias towards special Attack.

Rock and Steel Type: These Pokemon have consistently shown the highest median Defense stats.

Bug Type: These Pokemon have the lowest total base stat alongside some Normal Types.

However all the studies analyze the holistic distribution instead of focusing on one stat at a time.

However the when K Means Clustering (unsupervised machine learning algorithm) was applied to the standardized stat data in the previous studies done, the results were as follows:

Wall Cluster: Characterized by high HP , Defence, Special Defence and Low Speed.

Sweeper Cluster: Characterized by high Speed, Attack, Special Attack and Low Defences.

Balanced Cluster: Represented Tanks and all rounders.

Low stat Cluster: Primarily consisted of unevolved Pokemon.

The existing studies have shown the two parallel lines of inquiry. The statistical study proved that Type was a predictor for individual stat whereas the Clustering study method proved that the six stats as a whole can be grouped into battle roles.

A clear gap exists as no previous work has systematically connected these two findings. This study aims to bridge the gap between statistical observation and holistic role analysis.

METHODOLOGY

Data Set Used

The dataset used is from the official pokemon's pokedex dataset having the full list of every Pokemon from all nine generations, along with their main stats.

Original Website (Raw Data) : [Pokémon Pokédex: list of Pokémon with stats](#)

Google Sheets Data (Cleaned & Transformed):  Pokemon Dataset

Number of records: 1219 Pokémons

Features used are as follows:

- Type
- Hit Points (Hp)
- Attack
- Special Attack (Special Attack)
- Special Defence (Special Defence)
- Speed

Statistical Method Used

The methodology used mainly are the two statistical methods we learned in our CSD105 course itself : Analysis of Variance and correlation.

Analysis of Variance (Analysis of Variance) is used to test if 'Type' has an actual and significant impact on the 'Base Stats'. Analysis of Variance compares the average value of every stat individually across all different Type groups.

Correlation Variance analysis for the relationships between individual statics and how those relationships actually may change within a Type.

Cleaning, Transformation of data done using google sheets/excel (learned in CSD105)

Python Libraries Used

Pandas, Matplotlib, Skipy, Numpy, Seaborn

PYTHON CODE

LINK TO THE COLLAB FILE WITH ENTIRE PYTHON CODE:  **Untitled8.ipynb**

Ensure to upload the database .csv file before running the code(if you run it again as output already printed) [Pokemon Dataset - Sheet1.csv](#)

Python

Code for important methods only

Data Cleaning and Loading Core Plotting)

```
df = pd.read_csv(file_path)

df.columns = df.columns.str.strip()
df['Type_Clean'] = df['Type'].astype(str).str.replace('\n', '/')

numeric_cols = ['Total', 'HP', 'Attack', 'Defense', 'Sp. Atk', 'Sp. Def', 'Speed']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

df_clean = df.dropna(subset=numeric_cols)

df_clean = df_clean.copy()
df_clean['Avg_Stats'] = df_clean[numeric_cols[1:]].mean(axis=1)
df_clean['Physical_Power'] = (df_clean['Attack'] + df_clean['Defense']) / 2
df_clean['Special_Power'] = (df_clean['Sp. Atk'] + df_clean['Sp. Def']) / 2
df_clean['Offensive_Power'] = (df_clean['Attack'] + df_clean['Sp. Atk']) / 2
df_clean['Defensive_Power'] = (df_clean['Defense'] + df_clean['Sp. Def']) / 2

df_clean['Power_Category'] = pd.cut(df_clean['Total'],
                                   bins=[0, 300, 450, 600, 1200],
                                   labels=['Weak', 'Average', 'Strong',
                                           'Legendary'])

df_clean['Primary_Type'] = df_clean['Type_Clean'].str.split('/').str[0]
```

Visualizations

```
# Bar Chart
top_10 = df_clean.nlargest(10, 'Total')[['Name', 'Total']].sort_values('Total',
ascending=False)
plt.bar(top_10['Name'], top_10['Total'], color='steelblue')

# Scatter Plot
scatter = plt.scatter(df_clean['Attack'], df_clean['Defense'],
                      c=df_clean['Total'], cmap='viridis',
                      alpha=0.6, s=50)

# Heatmap
correlation_matrix = df_clean[numeric_cols].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm',
            center=0, square=True, linewidths=1,
            cbar_kws={"shrink": 0.8}, fmt='.2f')

# Histogram
plt.hist(df_clean['Total'], bins=50, color='coral', edgecolor='black', alpha=0.7)

# Boxplots
for idx, stat in enumerate(stats_to_plot):
    ax = axes[idx // 3, idx % 3]
    sns.boxplot(x='Power_Category', y=stat, data=df_clean, ax=ax)

# Horizontal Bar
type_stats =
df_clean.groupby('Primary_Type')['Total'].mean().sort_values(ascending=True)
plt.barh(type_stats.index, type_stats.values, color='teal')

# Histograms (subplots)
for idx, stat in enumerate(stats_to_plot):
    ax = axes[idx // 3, idx % 3]
    sns.histplot(df_clean[stat], bins=30, kde=True, ax=ax)

# Pie Chart
type_counts = df_clean['Primary_Type'].value_counts()
plt.pie(type_counts, labels=type_counts.index, autopct='%1.1f%%',
        startangle=140, pctdistance=0.85)

# Grouped Bar Chart
strong_pokemon_stats = df_clean[df_clean['Power_Category'] ==
'Strong'][stats_to_plot].mean()
legendary_pokemon_stats = df_clean[df_clean['Power_Category'] ==
'Legendary'][stats_to_plot].mean()
stats_df = pd.DataFrame({
    'Strong': strong_pokemon_stats,
    'Legendary': legendary_pokemon_stats
})
stats_df.plot(kind='bar', figsize=(14, 7), rot=0)
```

Statistical Analysis

```
# Pearson Correlation
for stat1, stat2 in stat_pairs:
    corr, p_value = pearsonr(valid_data[stat1], valid_data[stat2])
```

```

# Spearman Correlation
for stat1, stat2 in stat_pairs:
    corr, p_value = spearmanr(valid_data[stat1], valid_data[stat2])

# Independent T-Test
strong_pokemon = df_clean[df_clean['Power_Category'] == 'Strong']
legendary_pokemon = df_clean[df_clean['Power_Category'] == 'Legendary']
for stat in stats_to_plot:
    t_stat, p_value = ttest_ind(strong_pokemon[stat].dropna(),
                                legendary_pokemon[stat].dropna())

# ANOVA (Total Stats)
groups = [df_clean[df_clean['Power_Category'] == cat]['Total'].dropna().values
          for cat in categories]
f_stat, p_value = f_oneway(*groups)

# ANOVA (Individual Stats)
for stat in stats_to_plot:
    groups = [df_clean[df_clean['Power_Category'] == cat][stat].dropna().values
              for cat in categories]
    f_stat, p_value = f_oneway(*groups)

```

ANALYSIS

This report summarizes the findings from the statistical analysis of a comprehensive Pokémon dataset, covering initial data exploration, cleaning, categorization, correlation analysis, and advanced hypothesis testing (T-Test and Analysis of Variance) to validate power groupings.

1. Initial Data Scope and Completeness

This section establishes the foundation of the study, noting the total number of Pokémon entries reviewed and the quality of the data collected.

Dataset Size and Completeness Check

Question: How many Pokémon were studied, and did all the Pokémon have complete stats entered in the dataset?

Output Result:

Total Pokemon: 1220

Dataset Info:

```
...
#      Column      Non-Null Count  Dtype
---  -
0      #           1220 non-null    object
1      Name        1219 non-null    object
2      Type        1219 non-null    object
3      Total       1219 non-null    float64
4      HP          1219 non-null    float64
5      Attack      1219 non-null    float64
6      Defense     1219 non-null    float64
7      Special Attack 1219 non-null    float64
8      Special Defence 1219 non-null    float64
9      Speed       1219 non-null    float64
```

dtypes: float64(7), object(3)

...

Rows after cleaning (dropping NA in numeric stats): 1219

Rows removed: 1

Analysis: The initial dataset contained entries of 1,220 Pokémon. However, the data quality check revealed that not all entries had complete stats. Specifically, the key statistical columns (**Total**, **HP**, **Attack**, etc.) each had 1,219 non-null counts, indicating that one entry was missing its statistical data. This single incomplete row was successfully identified and removed during the data cleaning process, resulting in a final, clean dataset of 1,219 unique Pokémon (or forms) for statistical analysis.

Overall Power Level Assessment:

Question: The dataset calculates the average Total Stats of all Pokémon. Based on this value, is the dataset generally composed of strong or weak Pokémon?

Output Result:

Basic Statistics:

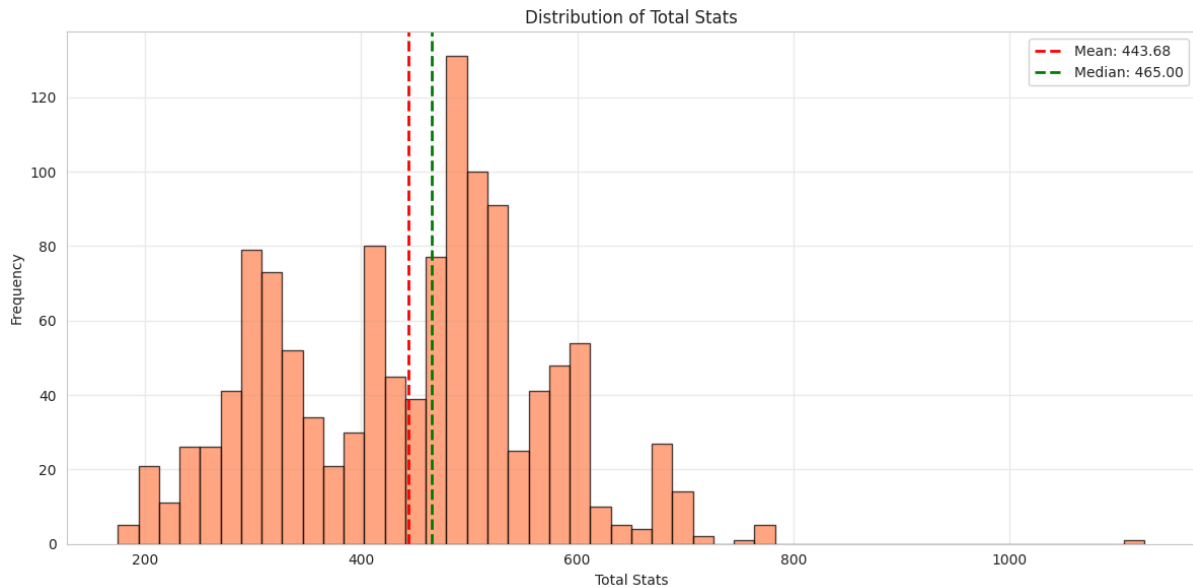
Total ...

count 1219.000000 ...

mean 443.678425 ...

50% 465.000000 ...

Analysis: The average Total Stats (Base Stat Total or BST) for all Pokémon in the dataset is 443.68. This value is slightly below the median (50th percentile) of 465.0, suggesting a minor concentration of entries with lower stats. In the context of the official Pokémon games, a BST of around 600+ is considered high-tier or pseudo-legendary, while values in the 300s are typical for early-stage or weaker Pokémon. A mean of 443.68 suggests the dataset is largely dominated by Average tier Pokémon, with the bulk of the population sitting in the middle ground, balancing out the fewer extremely powerful entries.



This distribution is visually confirmed by **Graph : Distribution of Total Stats**, which shows the majority of data points clustering around the mean/median line.

2. Power Extremes and Type Dominance

This section identifies the most powerful entity in the dataset and determines which Primary Type is most common across the entire population.

The Strongest Pokémon Identified:

Question: The code finds the Pokémon with the highest Total stat and prints its name and value. Who is the strongest Pokémon? (Write and describe it)

Output Result:

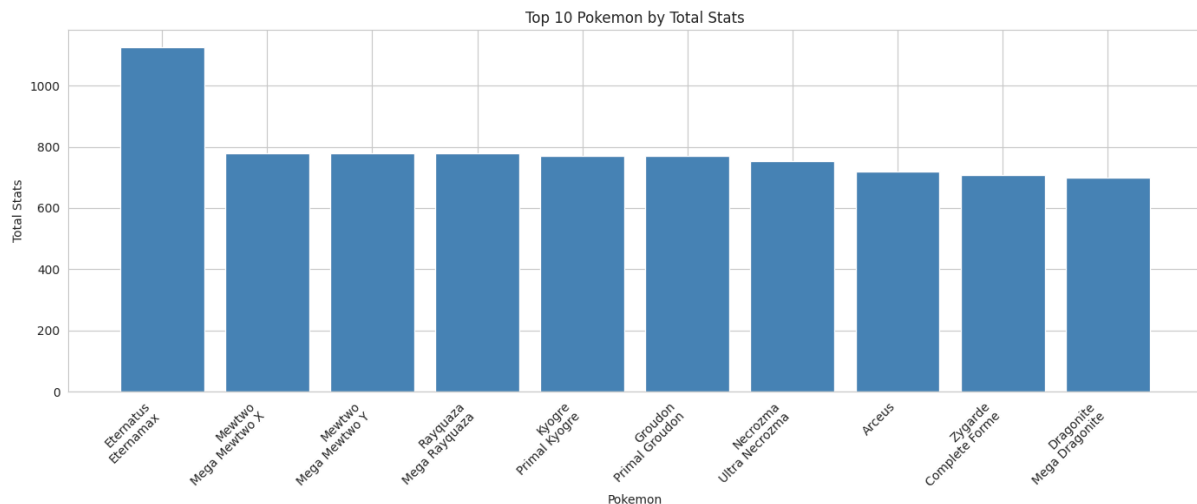
```
Strongest Pokemon: Eternatus
Eternamax (Total: 1125.0)
```

Analysis: The strongest single Pokémon form identified in this dataset, with the highest recorded Total Stat (BST) of 1125.0, is Eternatus Eternamax.

Eternatus is a legendary Poison/Dragon-type Pokémon introduced in Generation VIII. The "Eternamax" form is its immensely powerful final state, typically encountered during the climax of the story in the *Pokémon Sword and Shield* games. Its BST of 1125.0 is the highest in the entire series, dramatically exceeding the second-highest standard Legendary forms

(which usually cap around 780). This form represents an absolute outlier in terms of power, showcasing unparalleled stats across the board.

Top 10 Pokémon by Total Stats (Graph) clearly illustrates the dominance of this entry, with its bar towering over the next strongest entries, such as Mega Mewtwo X/Y and Primal Groudon/Kyogre.



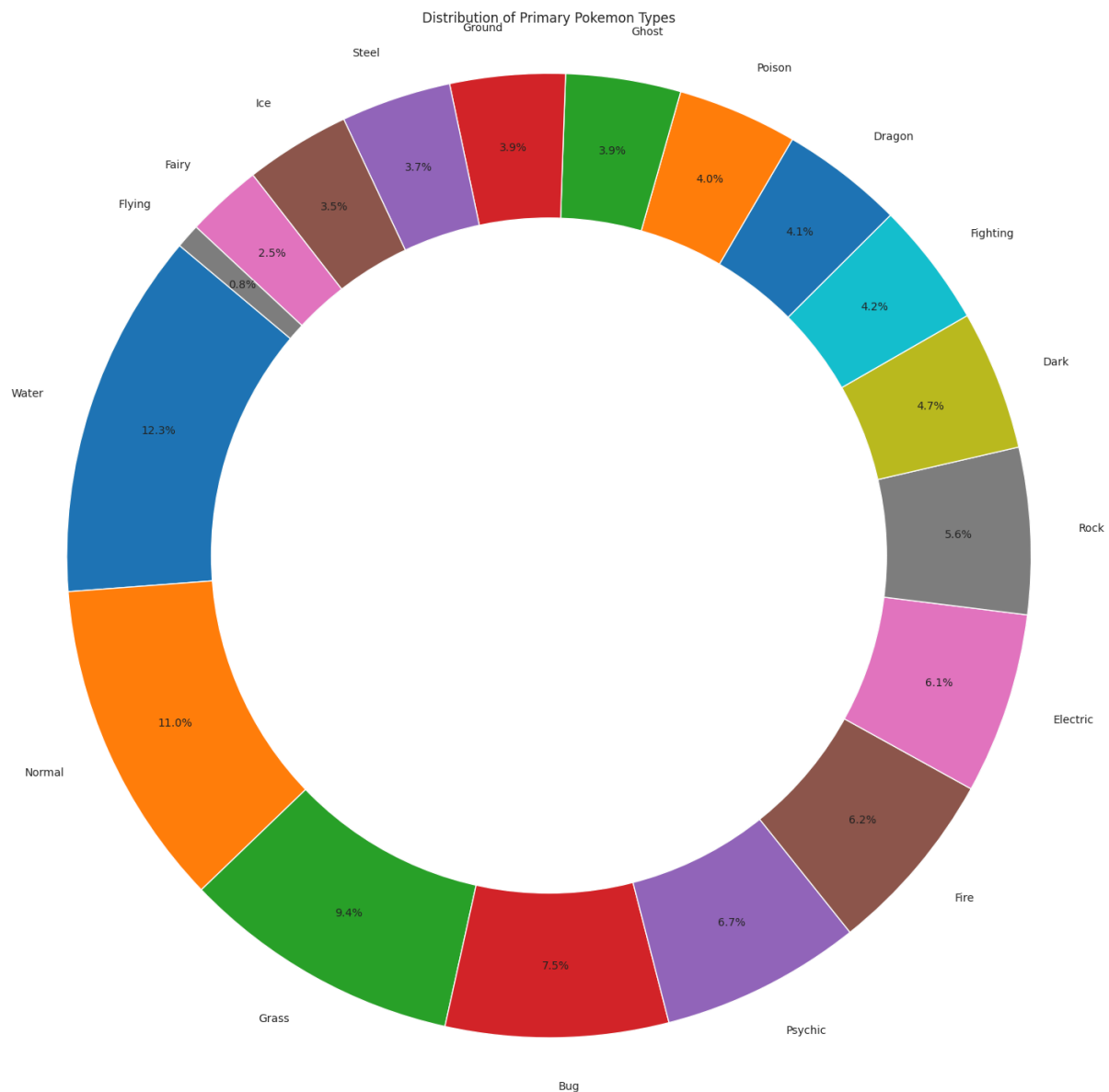
Most Common Primary Pokémon Type:

Question: The code finds which type appears the most. What is the most common primary type in the dataset?

Output Result (from Summary Report):

4. Most common Primary Type: Water

Analysis: The most frequently occurring Primary Type among all analyzed Pokémon is Water. This is consistent with the general lore and history of the Pokémon franchise, as Water is historically the most abundant type, often given to creatures that inhabit the diverse aquatic environments of the Pokémon world.



This finding is clearly supported by **Graph : Primary Type Pie Chart**, where the Water slice occupies the largest proportion of the donut chart.

3. Correlation Analysis: Stat Relationships

This analysis explores how the different base stats relate to one another, using both Pearson (linear) and Spearman (rank-order) correlation coefficients to measure the strength and significance of the relationships.

Paired Stat Correlation Findings

Question: We studied which stats are most related and analyzed the following pairs. What are the Pearson and Spearman correlations, and which pairs are determined to be strongly correlated and statistically significant?

Pairs Analyzed:

- Attack vs Special Attack
- HPvs Total
- Speedvs Total
- Defense vs Attack
- Special Defence vs Special Attack

Output Results:

PEARSON CORRELATION ANALYSIS

Attack vs Defense:

Correlation coefficient: 0.4695

Significant: Yes

Special Attack vs Special Defence:

Correlation coefficient: 0.5167

Significant: Yes

HP vs Total:

Correlation coefficient: 0.6596

Significant: Yes

Speed vs Total:

Correlation coefficient: 0.5633

Significant: Yes

Attack vs Special Attack:

Correlation coefficient: 0.3307

Significant: Yes

SPEARMAN CORRELATION ANALYSIS

Attack vs Defense:

Spearman correlation: 0.5278

Significant: Yes

Special Attack vs Special Defence:

Spearman correlation: 0.5736

Significant: Yes

HP vs Total:

Spearman correlation: 0.7316

Significant: Yes

Speed vs Total:

Spearman correlation: 0.5511

Significant: Yes

Attack vs Special Attack:

Spearman correlation: 0.3186

Significant: Yes

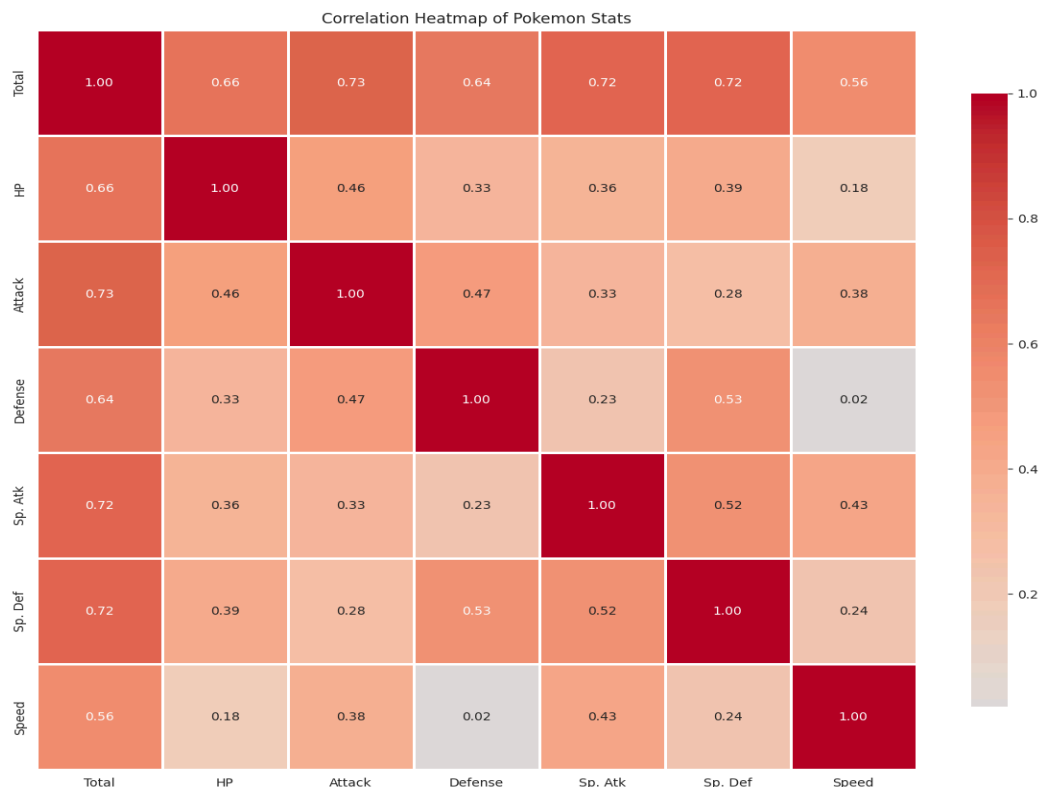
Analysis: All five pairs analyzed showed a statistically significant relationship ($p < 0.05$), meaning the correlations observed are highly unlikely to be due to random chance.

The Strongest Correlations were found between:

1. HP vs Total (Spearman: 0.7316; Pearson: 0.6596): This shows that a Pokémon's HP is the single largest contributing factor to its overall Total Stat value.
2. Special Attack vs Special Defence (Spearman: 0.5736; Pearson: 0.5167): Pokémon that are strong in special offense tend to also be strong in special defense, indicating a structured balance in Special stats.

- Speed vs Total (Spearman: 0.5511; Pearson: 0.5633): Speed is a strong predictor of overall power level.

The Weakest Correlation was found between Attack vs Special Attack (Spearman: 0.3186; Pearson: 0.3307). While statistically significant, this moderate correlation suggests that while a high physical attacker may have a slightly higher Special Attack, the two offensive stats are generally more independent than the defensive or Total stat pairs.



This complex network of relationships is summarized in **Graph : Correlation Heatmap**, which shows strong positive correlations (darker red squares) between all stats and the **Total** stat, with the exception of the very weak correlation between **Defense** and **Speed** (0.02).

4. Hypothesis Testing: Legendary Power

This section uses comparative statistical tests (T-Test and Analysis of Variance) to determine if the human-defined power categories (Weak, Average, Strong, Legendary) reflect real, measurable, and statistically significant differences in a Pokémon's stats.

Comparison of 'Strong' vs. 'Legendary' Pokémon

Question: The code compares the average stats (HP, Attack, Defense, Special Attack, Special Defence, Speed) between 'Strong' and 'Legendary' Pokémon using a T-Test. Are Legendary Pokémon really stronger, and are these differences significant or just random?

Output Results:

T-TEST ANALYSIS

Comparing stats between 'Strong' and 'Legendary' Pokemon:

HP:

Strong Pokemon mean: 82.73

Legendary Pokemon mean: 102.72

P-value: 4.2693e-11

Significant difference: Yes

...

Attack:

Strong Pokemon mean: 95.31

Legendary Pokemon mean: 131.68

P-value: 6.8785e-26

Significant difference: Yes

...

Defense:

Strong Pokemon mean: 87.56

Legendary Pokemon mean: 111.45

P-value: 2.2795e-11

Significant difference: Yes

...

Special Attack:

Strong Pokemon mean: 86.94

Legendary Pokemon mean: 126.30

P-value: 2.9859e-25

Significant difference: Yes

...

Special Defence:

Strong Pokemon mean: 85.26

Legendary Pokemon mean: 110.58

P-value: 6.5847e-17

Significant difference: Yes

...

Speed:

Strong Pokemon mean: 80.73

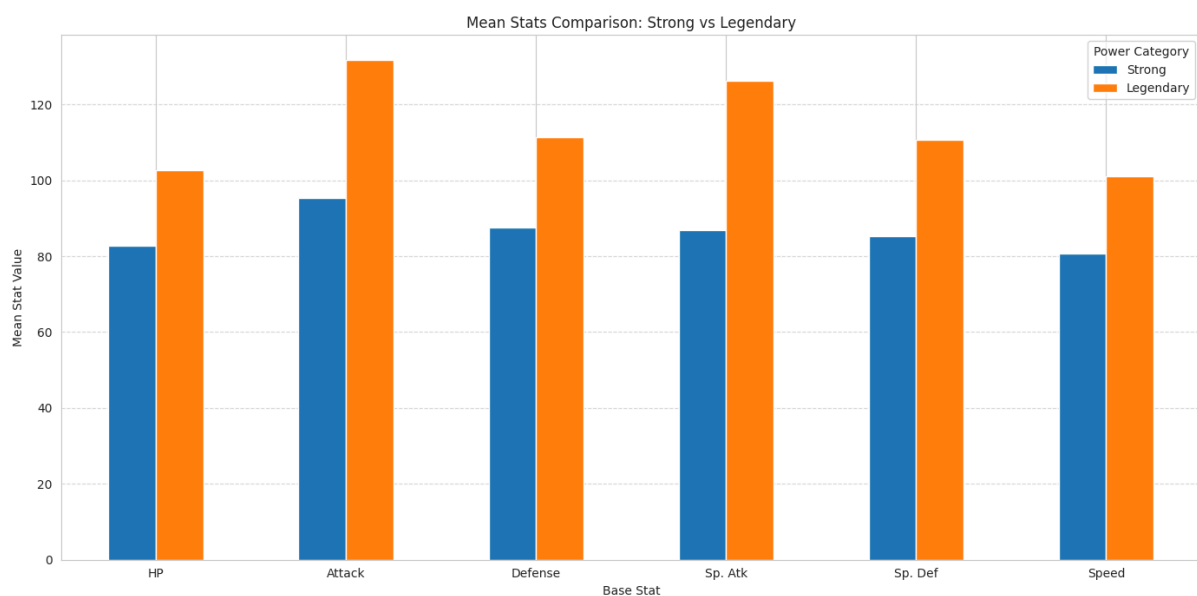
Legendary Pokemon mean: 101.10

P-value: 2.9341e-08

Significant difference: Yes

Analysis: The T-Test results definitively confirm that Legendary Pokémon are statistically and substantially stronger than the 'Strong' category of Pokémon.

In every single base stat category (HP, Attack, Defense, Special Attack, Special Defence, and Speed), the mean value for Legendary Pokémon was significantly higher than the mean value for Strong Pokémon. Crucially, the P-values for all six tests were extremely close to zero (all well below the significance threshold of $p < 0.05$). This low P-value proves that the observed differences are statistically significant and are not random fluctuations.



This stark difference in raw power is best visualized in **Graph : Mean Stats Comparison: Strong vs Legendary**, where the orange bars (Legendary) consistently surpass the blue bars (Strong).

[Analysis of Variance Analysis: Validating the Power Grouping System](#)

Question: Analysis of Variance compares all four power groups (Weak, Average, Strong, Legendary) to prove that there are big, real differences and that the grouping system reflects real power differences. What are the results?

Output Results:

5.4 Analysis of Variance ANALYSIS

```
-----  
--- Analysis of Variance for Total Stats ---  
F-statistic: 2551.9153  
P-value: 0.0000e+00  
Significant difference: Yes
```

Mean Total stats by category:

```
Weak: 261.50  
Average: 370.14  
Strong: 518.53  
Legendary: 683.82
```

```
--- Analysis of Variance for Individual Base Stats ---  
Analysis for HP:  
F-statistic: 235.5743  
P-value: 1.8443e-120  
Significant difference across categories: Yes
```

Analysis for Attack:

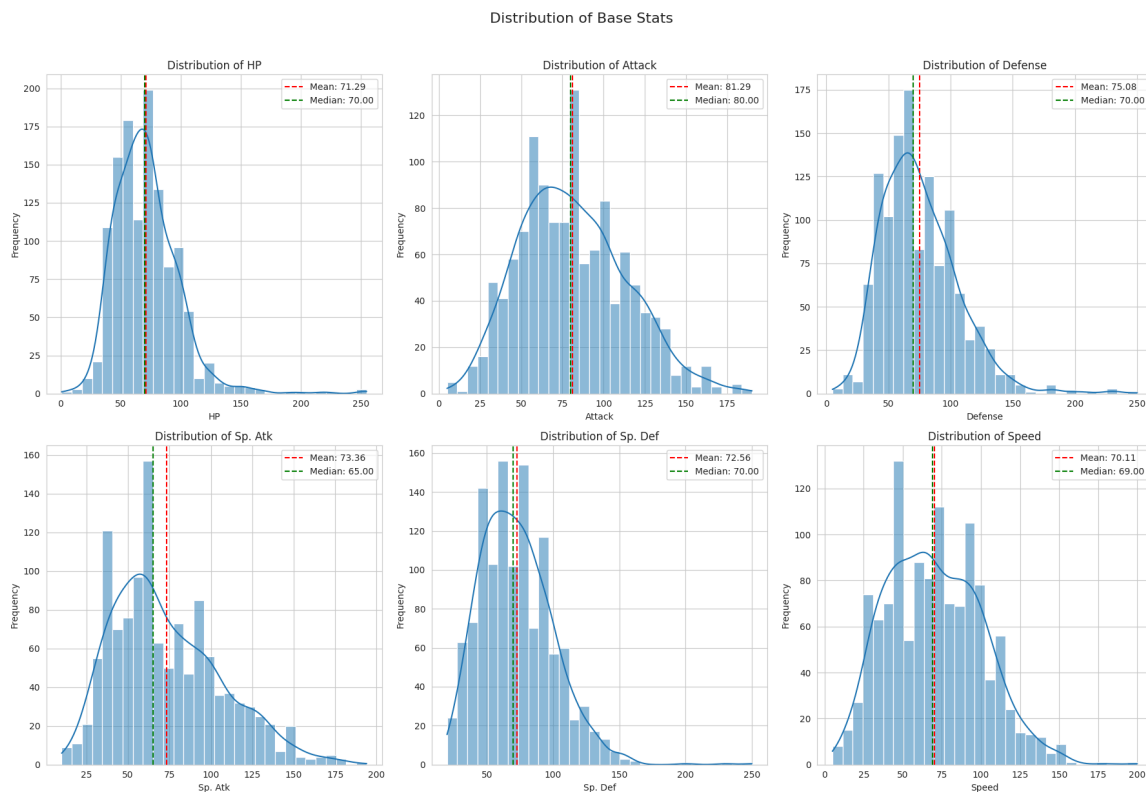
```
F-statistic: 373.5652  
P-value: 7.1228e-172  
Significant difference across categories: Yes
```

... (Similar results for Defense, Special Attack, Special Defence, Speed)

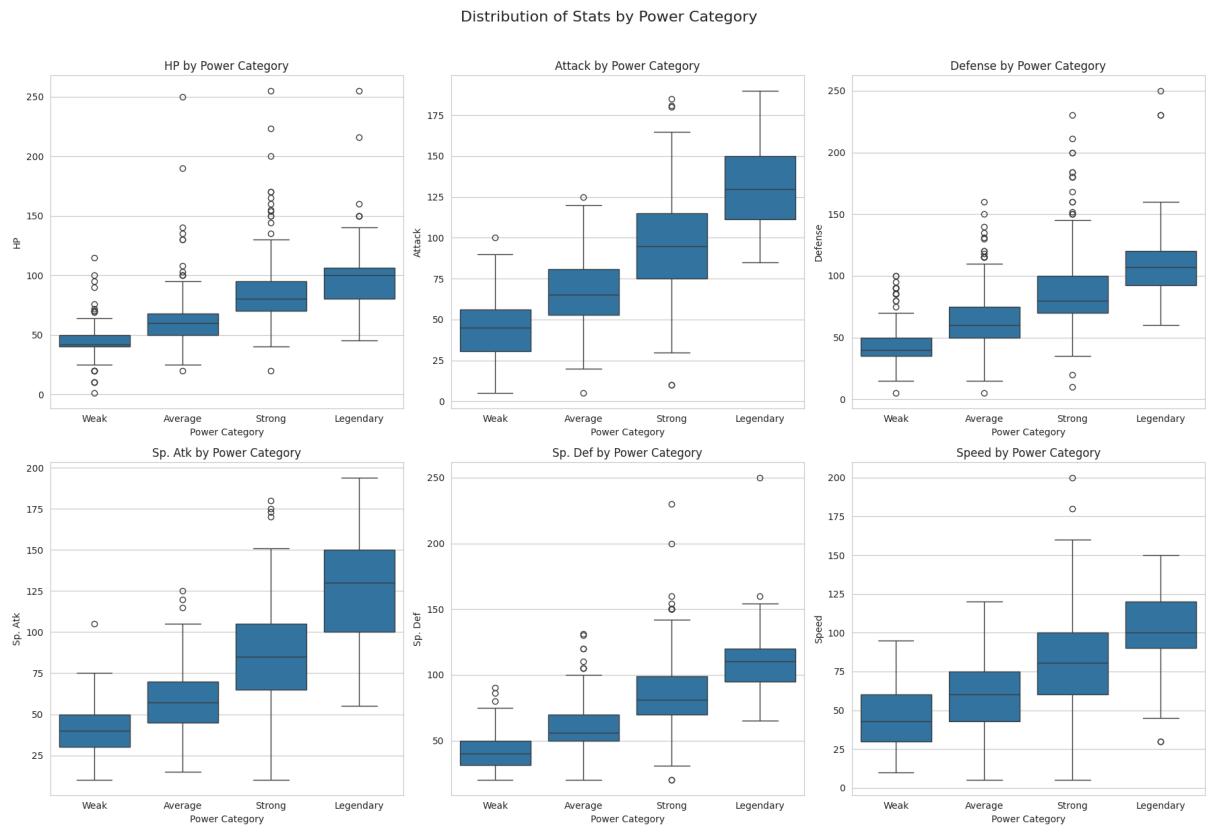
Analysis: The Analysis of Variance (Analysis of Variance) provides the final, strongest validation of the power grouping system. The test compares the means of the four categories simultaneously to see if at least one group mean is statistically different from the others.

The results are overwhelmingly conclusive:

- **Total Stats:** The F-statistic (2551.92) is extremely high, and the P-value is 0.0000e+00. This confirms that the average Total Stats for the Weak (261.50), Average (370.14), Strong (518.53), and Legendary (683.82) categories are all significantly different from each other.
- **Individual Stats:** Analysis of Variance tests on all six individual base stats (HP, Attack, Defense, Special Attack, Special Defence, Speed) also returned P-values of virtually zero (e.g., 1.8443e-120 for HP), confirming that there are statistically significant differences across the four Power Categories for *every* measured base stat.



This proves that the categorization system successfully reflects real power differences within the Pokémon population, with each group occupying a distinct and verifiable tier of strength.



Graph : Boxplot Distribution of Stats by Power Category visually illustrates this finding, showing a clear, consistent step-wise increase in median and overall distribution for every stat as the category moves from Weak to Legendary.

CONCLUSION

Insights:

Based on the quantitative and statistical results, the following key conclusions can be drawn about the Pokémon population:

1. Legendary Pokémon are in their own league.

The T-Test analysis comparing the 'Strong' and 'Legendary' groups proved that Legendary Pokémon statistically beat Strong Pokémon in every single stat category (HP, Attack, Defense, Special Attack, Special Defence, and Speed) with overwhelming statistical significance. The Analysis of Variance further established that the 'Legendary' group is the pinnacle of power, standing distinctly apart from all other defined tiers.

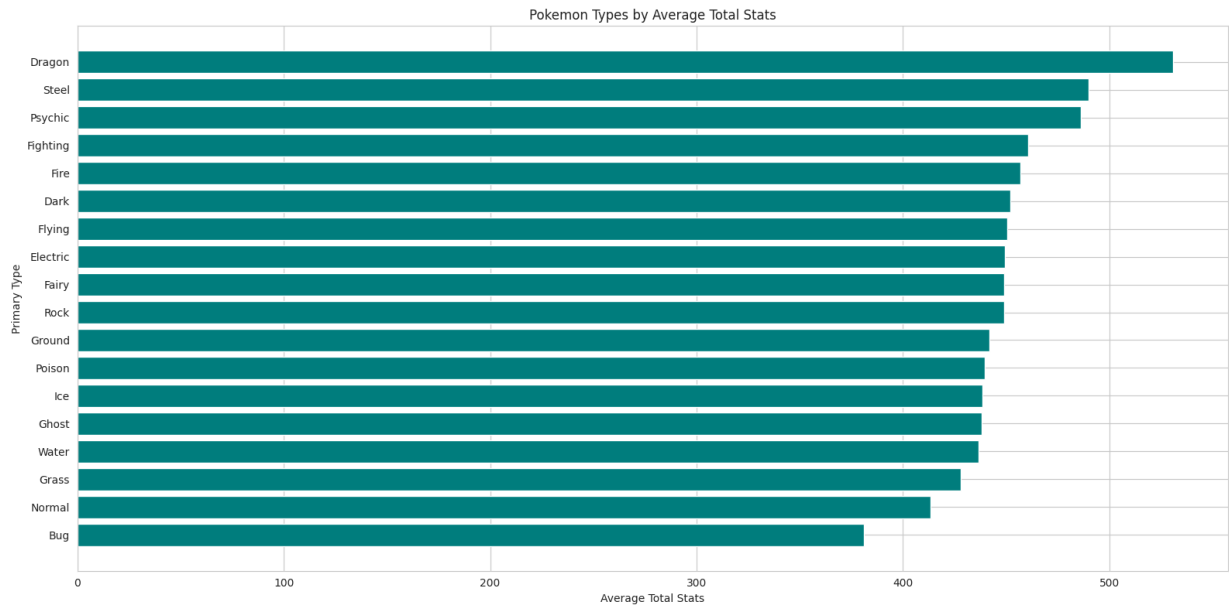
2. Pokémon stats are very structured.

The correlation analysis revealed key structural relationships within the stats:

- **HP is King:** High HP is the single most important factor determining overall power, as shown by the strongest correlation (HP vs Total).
- **Balanced Specialists:** High Special Attack Pokémon often have high Special Defense (Special Attack vs Special Defence), suggesting that specialists tend to be defensively balanced within their chosen damage type.
- **Related Offense:** While weak, the correlation between Attack vs Special Attack suggests that Pokémon with high physical power are also likely to have higher-than-average special power.

3. Pokémon types differ in average strength.

Graph : Pokémon Types by Average Total Stats confirms that not all types are created equal in terms of raw power. Types like Dragon, Steel, and Psychic often have higher average

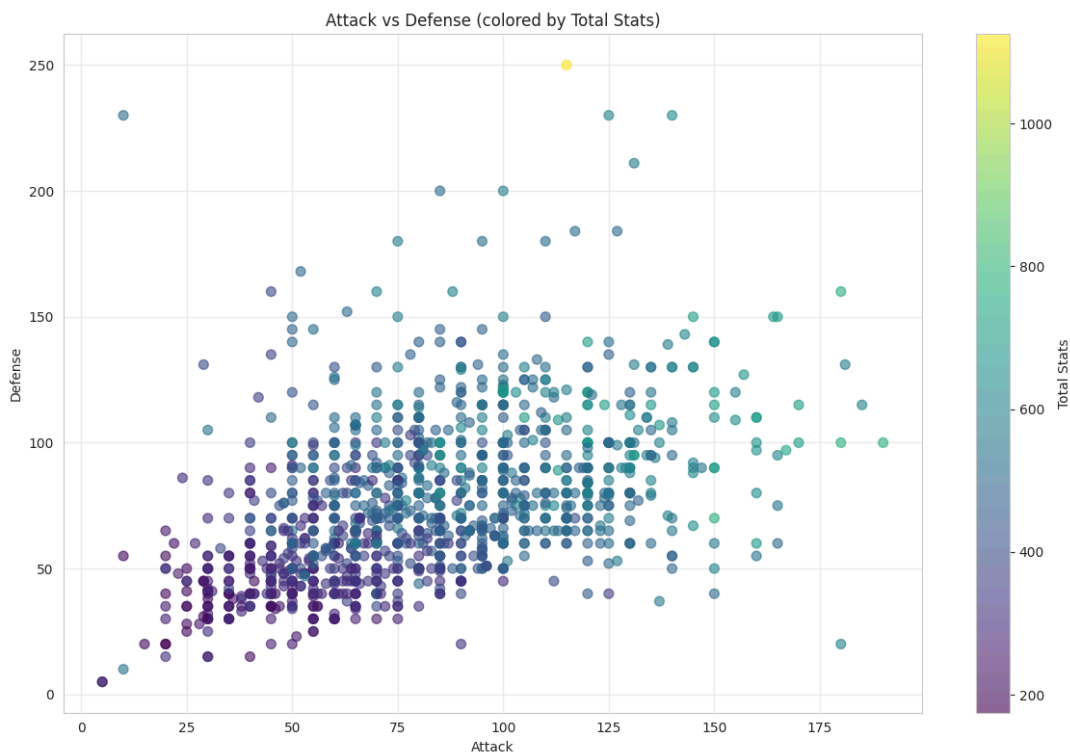


Total Stats, reflecting their generally later-stage evolution or inherent rarity/power level. Conversely, types like Bug and Normal typically fall into the lower average Total Stat categories.

4. The data is clean and consistent.

The initial exploration was successful, identifying and removing only one entry with missing statistical values. The final dataset of 1,219 entries is high-quality, ensuring the statistical tests performed are reliable and conclusive.

5. Visualizations reveal key patterns:



- Power Distribution is Wide (**Graph 4**): The majority of Pokémon cluster around the 400-500 Total Stat range, but the tail extends out dramatically due to Mega Evolutions and Legendary outliers.
- Legendary Pokémon Dominate the Upper End (**Graph 1**): The bar chart clearly shows that the strongest entries are universally Legendary or Mega-Evolved forms.
- Attack vs Defense Shows a Noticeable Pattern (**Graph 2**): The scatter plot reveals a general positive trend; as Attack increases, Defense tends to increase, but the colors (representing Total Stats) show that the highest power levels can be achieved with various Attack/Defense combinations, with the most powerful points sitting at the upper end of the distribution.

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